Perfume Recommendation System Based on Sentiment and Numerical Features: A Comparative Study of BERT, RoBERTa, and DistilBERT

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***Abstract*—The global perfume market has an annual value of 50 billion and still is expanding. While more and more perfumes are added in to the market, this makes fragrance selection for consumers difficult. Recommendation systems exist for this reason. But most existing fragrance recommendations only focus on note similarity while ignoring other factors such as performance or emotional values. This study proposes a method to close that gap. By using Aspect-Based Sentiment Analysis (ABSA) in Transformer-based models, this research is hoped to consider these subjective factors. Models like BERT, RoBERTa, and DistilBERT are considered and used in this study. They are tuned and trained on approximately 160,000 user reviews from *Fragrantica* across 740 perfumes. Other features such as longevity, sillage, and projection values from users are also used. Results show that BERT achieved best classification performance with weighted F1-Score of 96.18%. Meanwhile, in the recommendation system scoring, DistilBERT achieved the highest similarity score index by average of 0.84. Considering the lightweight aspect of DistilBERT, and the average results that differ only 1% from BERT, DistilBERT is chosen as the best model. Further analysis of the recommendation output showed that the model is sentiment-aware. This sentiment awareness can enhance future perfume recommendation systems in personalizing recommendations for users.**

Keywords— Recommendation System, Aspect-Based Sentiment Analysis (ABSA), Natural Language Processing (NLP), Transformer Models, BERT, RoBERTa, DistilBERT

I. INTRODUCTION

The global perfume industry is expanding at a very fast pace and has an annual value of USD 50 billion. This massive valuation of the market is directly connected to the number of products that are available. This is reflected in the presence of numerous digital platforms such as *Fragrantica*, *Parfumo*, *BaseNotes* that act as a community for the fragrance world. People can write reviews of perfumes that are available and post them in the forum. These reviews then form textual data that reflects users' preferences, and subjective evaluations such as longevity, sillage, and versatility which correlate to perfume's performance.

With numerous perfumes that are available and the vast number of reviews for each perfume, users can be overwhelmed by the information. Choosing between hundreds of thousands of perfumes that match their specific taste can be difficult. Most consumers tend to rely on recommendation tools when choosing fragrances. These systems typically depend on matching the scent components or notes. While seeming to be useful for base recommendation, It ignores personal taste and mood from users entirely [1].

Using models built on the Transformer architecture, such as BERT, RoBERTa and DistilBERT [2][3][4] in NLP can handle previous generation issues. These transformer models can understand deeper meanings, and subtle phrases between sentences. Because of this, they are perfect for examining the detailed and heavily nuanced text in perfume reviews.

Combining with Transformer architecture, Aspect-Based Sentiment Analysis (ABSA) will also be integrated in to the system. ABSA works by dividing sentence into specific features and analyzing sentiment for each aspect. It analyzes surrounding context and adjectives for better sentiment analysis. When used in fragrance review texts, it understands reactions to traits like how long they last, their projection, adaptability across occasions, cost versus quality, and suitability by season [5].

This study intends to close the gap from previous recommendation systems by using a new system. The proposed system is utilizing Transformer models with ABSA. These combinations are to make sure that every aspect of the perfumes, be reviews or scent profiles are to be accounted for. That will in turn generate relevant yet unique perfume recommendations for users.

II. RELATED WORKS

A. Natural Language Processing and Transformer Models

Older NLP models were built using statistical methods such as Bag-of-Words, and TF-IDF to categorize texts [2]. These methods treated words as isolated data points, which in turn fail to capture the bidirectional context that surrounds the word in a sentence. In perfume review texts this will mean that they will miss the nuance of the descriptions. For example, "fresh" when paired with "citrus" or "clean" will be interpreted differently.

To solve this, Vaswani et al. [2] introduced Transformer architecture in their study. This finding shifted NLP away from sequential recurrence toward self-attention mechanisms. This mechanism enables the model to process tokens in parallel while simultaneously capturing complex contextual relationships.

Transformer consists of an encoder–decoder structure, where each component is composed of stacked layers containing multi-head self-attention and feed-forward networks as can be seen by the illustration in Figure 1. The encoder processes the input sequence and learns contextual representations by assigning varying attention weights to different words. That then model the global dependencies across the sentence. The decoder generates output sequences by attending to both the encoder's output and previously generated tokens, using masked multi-head attention to prevent information leakage from future positions.

Using the Transformer architecture as its base, several pre-trained language models emerged. These language models are already trained by their respective datasets and capable with many different NLP tasks. The most popular is BERT (Bidirectional Encoder Representations from Transformers) which introduced bidirectional contextual understanding through Masked Language Modelling (MLM) and Next Sentence Prediction (NSP) [3]. There is also RoBERTa (Robustly Optimized BERT Pretraining Approach) that refined this process by removing the NSP objective, applying dynamic masking, and leveraging larger mini-batches and training corpora to enhance generalization [4]. Meanwhile, other model such as DistilBERT which is developed by Sanh et al. [5], employed knowledge distillation to compress the BERT model, reducing its size by 40% while retaining 96% of its performance. These ideas and results made Transformer-based models both scalable and efficient.

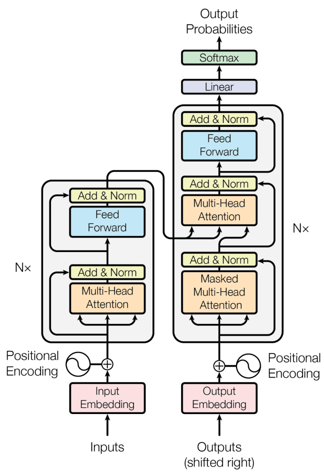

Fig. 1. Transformer Architecture (Vahsani A, et al. 2017)

B. *Sentiment Analysis and Aspect-Based Sentiment Analysis*

Sentiment Analysis is used to determine the emotional tone and the polarity of a text. This analysis will be used to know or reveal users' intentions and opinions against specific products. Previous sentiment classifiers used lexicon-based approach and traditional machine learning algorithms. Those algorithms include Naïve Bayes, Support Vector Machines (SVM), and Decision Tree [6]. Even though the models achieved good results and accuracy, they were still struggling with ambiguity and sarcasm that are available in some texts that are subjective like reviews.

To combat the context ambiguity the previous sentiment classifiers had, deep contextual models come as a solution. Particularly, Transformer-based architectures have significantly improved sentiment understanding. BERT outperform these traditional methods in extracting nuanced opinions. Leveraging self-attention and contextual embeddings, extraction of contexts from many sources be it formal or informal are possible.

ABSA focuses on separating words from sentences. This help by further evaluating adjectives and contexts surrounding a word. Making ABSA better and more precise than previous sentiment analysis model [4]. In perfume review texts, this mean dissecting reactions of users to scent profiles, such as citrusy or woody. Moods, vibes and atmospheric contexts also being considered. Whether perfume is better worn in summer or has clean vibes or better worn in night is relevant in the analysis. Transformer models handle this better than previous architectures, such as CNNs or RNNs [7]. Mainly because of Transformer models' atttention heads that gather models' attention where needed.

C. *Sentiment Aware Recommendation System*

Sentiment analysis boosts recommendation systems, pushing them past simple rating-based or collaborative methods. Previous models typically use similarity or user-item interaction matrices [8]. These approaches have flaws, such as often failing to capture the subjective and emotional aspects that actually determine a recommendation relevancy.

A study by Gheewala, S. et al. [8] used a hybrid approach to improve prediction of user satisfaction. Their system use NLP integratiion with colaborative filtering that has proved to be successful. Another study by Darraz et al. [9] that use Transformer model to extracting sentiments. These sentiments then will be used in a recommendation system, that they have proved to be more diverse and relevant.

But most of these sentiment based recommendation system studies are used on e-commerce, movies reviews [10]. These reviews are structured and is direct compared to perfume reviews. Perfume reviews are usually subjective, packed with emotions, and subtle phrases that make sentiment analysis more difficult. This research hope to close this gap by using ABSA with Transformer models.

D. *Rationale for Method and Models Selection*

Traditional sentiment classifiers such as Support Vector Machines (SVM) and Naïve Bayes struggle with nuances [6]. They often miss the metaphors and subjective flair typical of fragrance reviews. This limitation necessitated the shift to Transformer-based architectures [2].

BERT, RoBERTa, and DistilBERT are selected because of their ability of balancing accuracy with efficiency. BERT is the base of this technology by using bidirectional learning. RoBERTa builds on BERT base, and add more dataset training to be done. Which resulted in more optimized output. DistilBERT is the lighter version of BERT. Only having 40% of BERT's size but still retain 96% of BERT's capabilites. This makes it ideal for practical deployment becase of its reduced computational costs.

Evidence from Perikos and Diamantopoulos [7], alongside Karabila and Darraz [9], have shown that Transformer-extracted embeddings outperform standard collaborative filtering in both accuracy and user satisfaction. These findings support our decision to integrate ABSA-based sentiment modeling directly into a cosine similarity recommendation engine. We can then use this new method as a way to handle the emotional aspect of selecting scents through integrating the sentiment analysis with the ranking process.

III. METHODOLOGY

This chapter describes the overall methodology used in this study. The figure of the system is illustrated below in Fig 2.

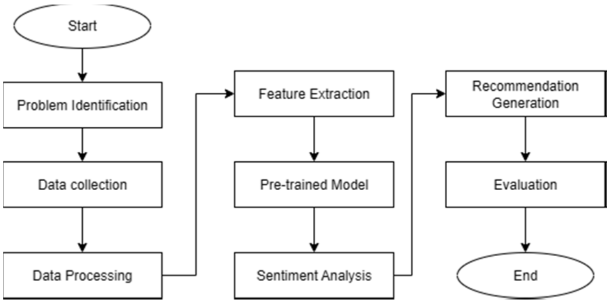

Fig. 2. Research Workflow

Firstly, a problem description stage is undertaken in which the goal of developing an improved perfume recommendation by utilizing sentiment-based features. User reviews and perfume-related metadata are collected from *Fragrantica* (https://www.fragrantica.com/). After data is collected, the data will be preprocessed. The data will be cleaned, normalized, and structured in terms of features so as to prepare it for model training.

During the feature extraction phase, the textual reviews are translated to numerical representations using pre-trained transformer models (such as BERT, RoBERTA, and DistilBERT) to be utilized in the research. These transformer models are used to perform sentiment analysis on the reviews to obtain both the emotional tone and aspect-level sentiment of each review.

Last, the sentiment embeddings are sent to the recommendation generation module, where the Cosine Similarity algorithm is utilized to develop personalized perfume recommendations. Then, the efficacy of the system is assessed via sentiment classification metrics and performance metrics for perfume recommendations, thus completing the methodology workflow.

A. Dataset

TABLE I. SAMPLE OF A SELECT FEATURES OF PERFUME DATASET.

Perfume Name	Brand	Rating	Example Review
Terre d'Hermès	Hermès	4.28	"Earthy orange peel... deeper in the rain."
Aventus	Creed	4.45	"Powerful, masculine scent ... and longevity."
Voyage	Nautica	3.92	"Great scent for the gym, ... good fruity smell."

The dataset contains roughly 160,000 user reviews accross 740 unique perfumes. Some of the features are perfume name, designer brand, release year, and gender classification (men, women, or unisex). Beside static labels, we also incorporated the user voting data to gauge each perfume's performance. Users rate attributes on specific intensity related to longevity, sillage, and projection. These attributes are rated on an intensity scale from weak to strong for projection, very weak to long-lasting for longevity, and intimate to enormous for sillage. Most importantly, the dataset pairs these structured metrics with unstructured textual reviews which can be seen in short form in Table I.

B. Data Preprocessing

The preprocessing stage starts with text cleaning and normalization. To ensure consistency, reviews are converted to lowercase, and HTML tags, emojis, punctuation, and extra spaces are removed. Reviews with fewer than five words or written in non-English languages are excluded. The remaining reviews are then tokenized using the *WordPiece* tokenizer corresponding to each Transformer model (BERT, RoBERTa, or DistilBERT). The tokens are then converted into indices, padded or truncated to a maximum length of 256, and encoded into embeddings.

After tokenization, stopwords are removed using the NLTK English stopword list, and lemmatization is applied to reduce words to their base forms. Numeric attributes, such as ratings and vote counts, are normalized to enable integration with textual embeddings. The final preprocessed dataset thus consisted of both cleaned textual reviews and standardized numerical features for model fine-tuning.

C. Model Architecture

Pre Trained Language Models: BERT, RoBERTa, and DistilBERT, are fine-tuned for Aspect-Based Sentiment Analysis. These models generate contextual embeddings from textual input, which will be processed by a classification head to determine sentiment polarity for each aspect (positive, neutral, or negative). The models' configuration that are going to be used are presented in Table II.

TABLE II. FINE-TUNING PARAMETERS FOR TRANSFORMER-BASED ABSA MODEL TABLE

Parameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-5}
Batch Size	16
Epochs	5
Max Token Length	256
Loss Function	Cross-Entropy
Framework	PyTorch & Transformers

D. Sentiment Embedding and Recommendation Process

Figure 3 shows the overall workflow of this study. It summarizes the main steps of the system, from data cleaning to the generation of personalized perfume recommendations.

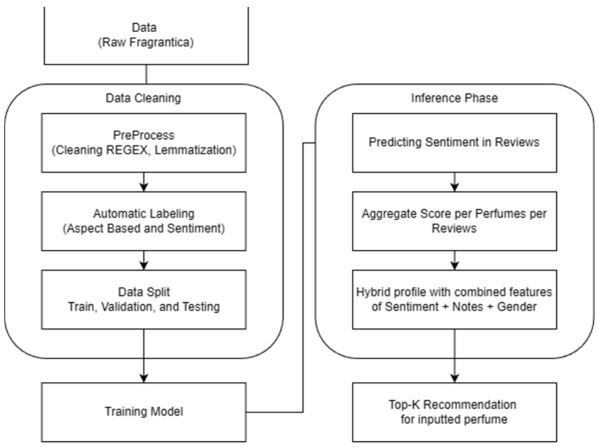

Fig. 3. Systematic Flow

Illustrated in Figure 3, after fine-tuning, each Transformer model generates aspect-specific sentiment embeddings that capture users' emotional responses and experiences related to different perfume attributes. These features contain user votes of longevity, sillage, projection, and value of the perfumes.

Continuing the process, each features will be weighted by relevance in the review. From sentiment polarity to voting data attributes such as longevity, proojection, and sillage will also be integrated. The system will then rank perfumes from the dataset by smilarity score of the features combined. Output will be Top-K list based on this similarity score.

E. Evaluation Methodology

Evaluation of the system will be divided into two main segments. Sentiment classification as the first, will be for ensuring model understand meaning of the review texts and give them polarity that is best suited. Second will be Recommendation relevance is for how well the model utilize the nuances of the perfumes and combining it with similarity score and generating a relevant list of recommendations.

For sentiment classification, standard performance metrics were used: Accuracy, Precision, Recall, and F1-Score. Weighted-F1 scores were prioritized because of the imbalanced nature of review datasets, which is usually skewed into a specific class. This is to ensure neutral and negative classes were fairly represented. Confusion Matrix are also used to visualize specific misclassification trends. Helping identify problems that the models might face.

For recommendation module, live interaction metrics like MAP or NDCG are not included in this research. Instead, the system will be validated using Similarity Score Analysis. By mapping the cosine similarity distribution within the Top-K (K=5) results, we gauged the model's confidence in matching the input profile's sentiment and olfactory features. This analysis include comparing the distribution of the cosine similarities of the Top-K output between the models. From there, qualitative analysis will also be done by looking at the Top-K output. System are hypothesized to be able to recommend the flanker version of the Top-K perfume higher, while also recommend perfumes of the same gender in Top-K. For example, when the user inputted a perfume that is targeted for men, the system will hopefully ignore perfumes that are targeted for women and vice versa while still recommending perfumes that are unisex.

IV. Results and Discussion

This chapter shows the performance of the system, the sentiment classifier and also the recommendation module. BERT, RoBERTa, and DistilBERT are compared to show which are the best to be integrated in the system.

A. Sentiment Classification Performance (ABSA)

Using Accuracy, Weighted Precision, Weighted Recall, and Weighted F1-Score as the measurement, each model are assessed how accurately they differentiate between positive, neutral, and negative. Table III details these results.

TABLE III. Performance Metrics for ABSA Models

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)
BERT	**96.91%**	**96.20%**	**96.22%**	**96.18%**
RoBERTa	96.15%	95.42%	95.76%	95.09%
DistilBERT	96.08%	95.43%	95.72%	95.15%

All three Transformer models achieved strong performance, with accuracy values consistently exceeding 96%. This indicates their ability to effectively capture the nuanced and often metaphorical language commonly found in perfume reviews. Among the evaluated models, BERT demonstrated the best overall performance, achieving the highest weighted F1-Score of 96.18%.

DistilBERT also produced competitive results, with weighted F1-Score of 95.15%, which is comparable to the larger models. Notably, DistilBERT slightly outperformed RoBERTa, which achieved weighted F1-Score of 95.09%. These findings suggest that DistilBERT is a viable lightweight alternative for deployment, as it maintains performance close to BERT while offering improved computational efficiency.

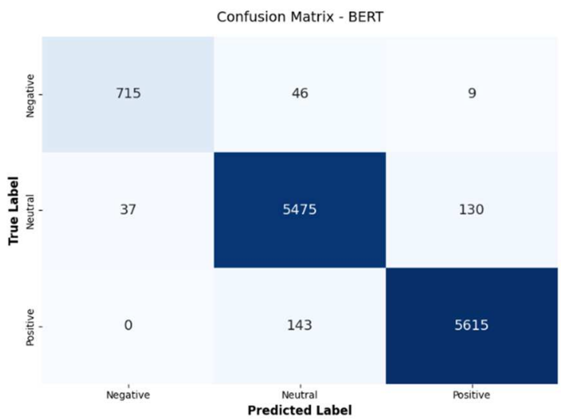

Fig. 4. Confusion Matrix for BERT

Figure 4 shows the confusion matrix for the best performing model (BERT). It shows that BERT performed well across all sentiment classes. BERT correctly classified 5,615 positive reviews and is also accurate in identifying the minority class (negative), with 715 correct predictions. Small misclassification were also made for negative labeled as positive (9) and positive labeled as negative (0). Meaning the model will not recommend products that are disliked and will not reduce the score of 'good quality' perfumes.

BERT also showed good result in classifying neutral reviews, with 5,475 correct predictions. This suggests that the model is sensitive and able to differentiate neutral and positive sentiment. Which then help the recommendation system distinguish between perfumes that are acceptable and those that are genuinely preferred.

B. Recommendation System Performance

The recommendation system will be run by three different models and compared. To examine the versatility and semantic accuracy across different fragrance categories, three case studies were used: *Terre d'Hermès (male), Coco Mademoiselle (female), and CK One (unisex).*

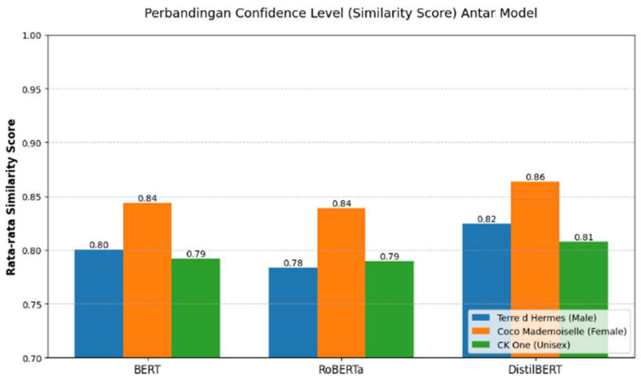

Fig. 5. Comparison Chart of Average Similarity Scores between Models

As shown in Figure 5, DistilBERT produced slightly higher similarity scores than BERT and RoBERTa across all test categories. In the *Coco Mademoiselle* (female) case study, Three of the models achieved their respective highest similarity score compared to other case studies. But, DistilBERT achieved the highest average similiarity of 0.86 while BERT and RoBERTa reached average scores of 0.84.

The small differences in similarity scores among the models, ranging from 0.01 to 0.04, suggest that all three Transformer models are capable of capturing semantic features within the perfume domain. However, DistilBERT's slightly higher scores indicate more stable performance for fragrance profile matching.

To further validate the system's semantic relevance, the Top-5 recommendations produced by DistilBERT were analyzed and are presented in Table IV.

TABLE IV. TOP-K RECOMMENDATION RESULT FOR DISTILBERT.

Terre d'Hermès	***Coco Mademoiselle***	**CK One**
Terre d'Hermes Eau Intense Vetiver for men	Psychedelic Love for women and men	Versace Versense for women
Missoni Wave for men	Kenzo Jungle L'Elephant for women	Aqua Allegoria Herba Fresca for women and men
Burberry Touch for men	Narciso Poudree for women	Cacharel Noa for women
L'Homme Idéal for men	Good Girl Blush for women	L'Eau d'Hiver for women and men
L'Eau d'Hiver for women and men	Libre Le Parfum for women	Chloé Nomade for women

In the first case study, the system chooses *Terre d'Hermes Eau Intense Vetiver for men* as the first pick. This indicate that the system prioritizes flankers, perfumes that share the same attribute with the main perfume. Labeled as "Intense" that generally get better performance, the system takes that into account and put it high. The results demonstrate that the system is able to deliver recommendation not only based on the metadata (perfume notes) but with sentiment in mind.

In the second case study, the system recommends *Psychedelic Love* for the first pick as the top match for *Coco Mademoiselle*. Though sharing patchouli-floral characteristics, *Psychedelic Love* is a premium niche unisex fragrance from *Ignitio*. This recommendation is likely because of the 'premium' quality of the fragrance itself. But with these result, it is known that the system also rate better quality perfumes higher than others.

Arguably the most important, the third case highlights the system's capacity to recommend different marketed genders. Usually gender labeling on perfumes are just for marketing purposes, most of the time users can wear fragrances across gender lines. Can be seen from the first pick of the recommendation which is *Versense*, that is marketed for women. This recommendation mostly because citrus-aromatic profile of the two fragrances instead of their gender profiles . Supporting this, the rest of the picks are also from different marketed genders.

C. Discussion

From the results gathered, BERT achieved the best performance in sentiment classification task, and is great because of the low misclassification trends. Generally, this will also result in better recommendation generation. But, with the experiments of limited test cases, DistilBERT actually get the highest similarity scores. Keeping in mind that DistilBERT is more lightweight than BERT, which means faster loading time. DistilBERT also got close result with BERT, only a 1% margin of F1-Score. With these in mind, DistilBERT will be chosen as the best model.

Recommendation outputs of the system also matches the hypothesis of the system. Alongside matching the metadata of the perfumes, System is also able to detect flankers and recommend them higher than the others while noting the 'strong' performance sentiment. System also rates scent quality highly and able to integrate different gender profiles in the recommendation output.

V. CONCLUSION

A. Conclusion

BERT showed that it is capable of understanding complex sentiments in perfume reviews. It got minimal misclassification errors showing good potential for recommendation systems. DistilBERT also showed very promising results from the similarity scores while being lighter than BERT. DistilBERT achieved close results compared to BERT with a 1% difference in weighted F1-Score. With all these advantages, DistilBERT is chosen to be the optimal model for these experiment.

More analysis are done for the output recommendations. The system is successful in recommending relevant products. Comparing similarity metadata of the fragrances, and also focus on flankers, scent quality and sentiment awareness. System also capable of recommending different marketed gendered perfumes, and in turn enhancing perfume discovery.

B. Limitations and Future Work

Even with all these promising results, this research has several limitations. Especially in dataset scope. The dataset is only consisted of 740 unique perfumes. This may affect the system selection of recommended perfumes, with a bias of recommending the most popular perfume which often have more reviews. The dataset is also static, and have cold start problem where the system doesn't work well for newer perfumes that have lesser reviews and lack community interactions and feedback.

Furthermore, the current recommendation system relies primarily on similarity of the perfume notes and sentiment-derived features, excluding other influential decision factors such as price, brand reputation, and packaging aesthetics. This is also supported by not having real world tests and other qualitative feedback.

Future work should address these limitations by using dataset that are larger and have more unique perfumes and reviews. Including users' profile in collaborative-filtering system will also be better for handling the cold-start problem. Using real world surveys and experts' feedback will solidify the recommendation output further. Should also consider ease of use for users by giving explanations for the recommendation output list.

Author Contributions

Widjaja. A. W. was responsible for the research design, literature survey, data collecting and processing, experimental execution, analysis, and initial manuscript preparation. Anggreainy. S. M. guided the conceptual direction, supervised the project, and refined the manuscript through critical editing. Both authors take responsibility for the study's integrity and have approved the final submission.

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